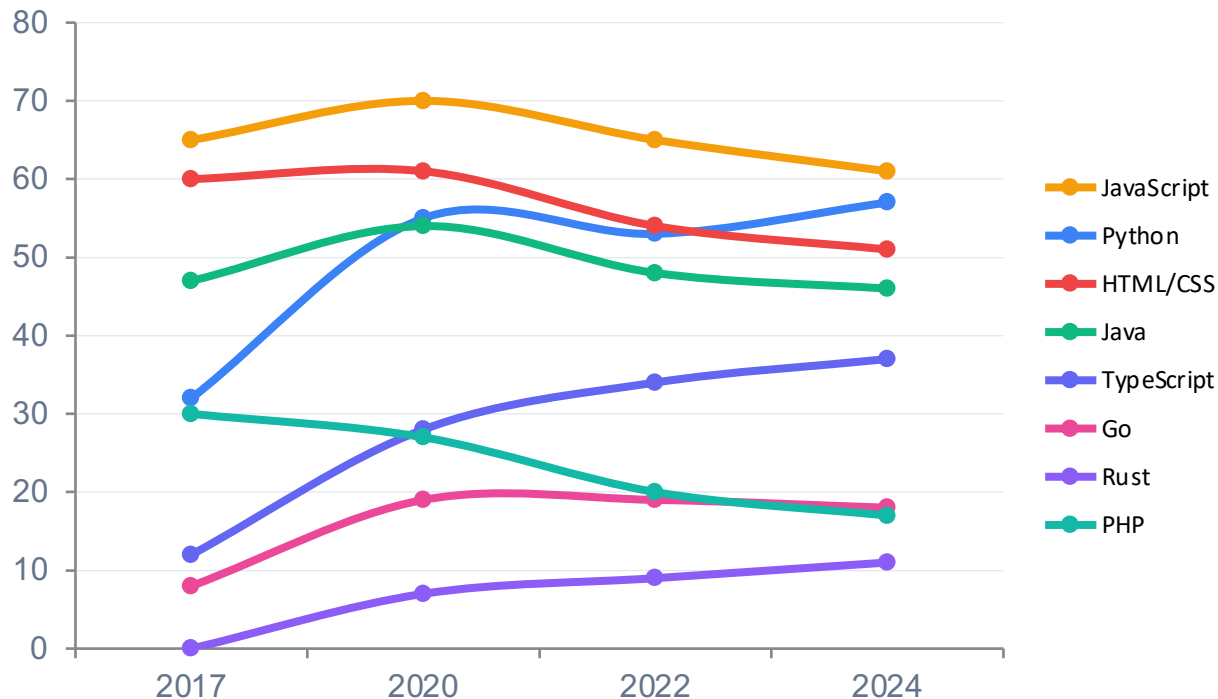


STATE OF DEVELOPER ECOSYSTEM 2024

Language Trends, Satisfaction, and Strategic Shifts

Key insights from JetBrains' survey of 23,262 developers worldwide.

Python Closes the Gap: 8 Years of Language Usage Shifts



Key Shifts

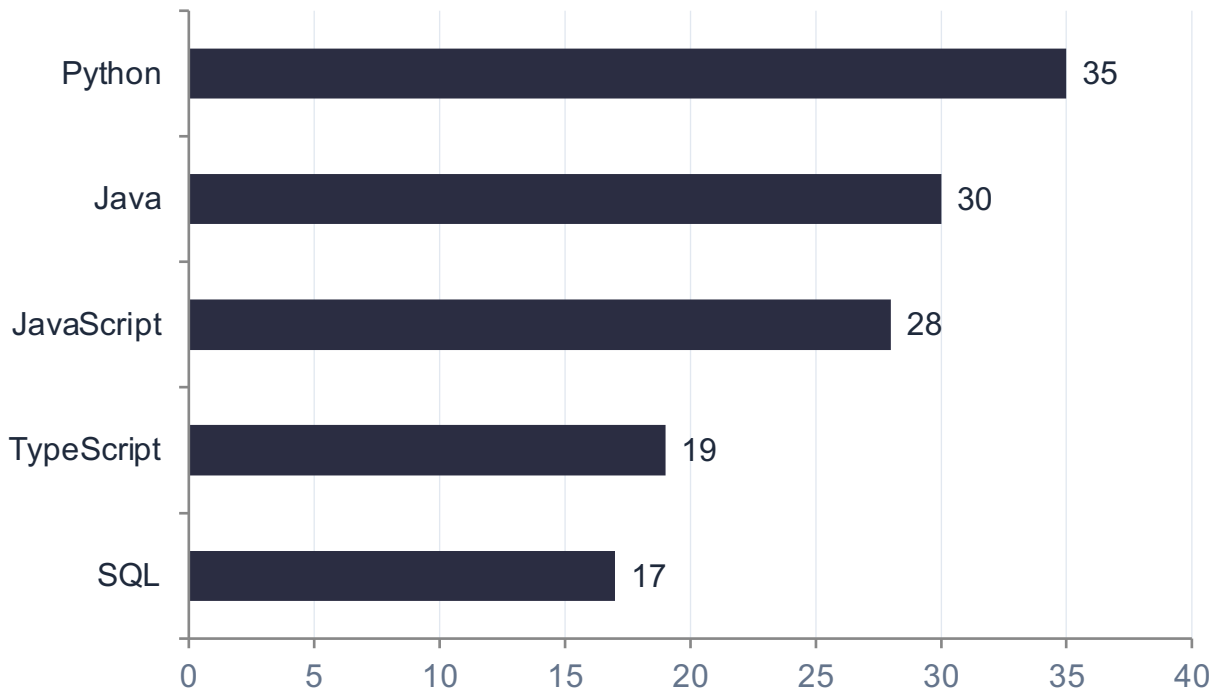
JavaScript peaked at 70% in 2020 before declining to 61%.

Python rose from 32% to 57%, nearly closing the gap with JS.

TypeScript tripled its share from 12% to 37%.

PHP saw a significant drop from 30% to 17%.

Python Is Now the #1 Primary Language at 35%



Ecosystem Lock-in

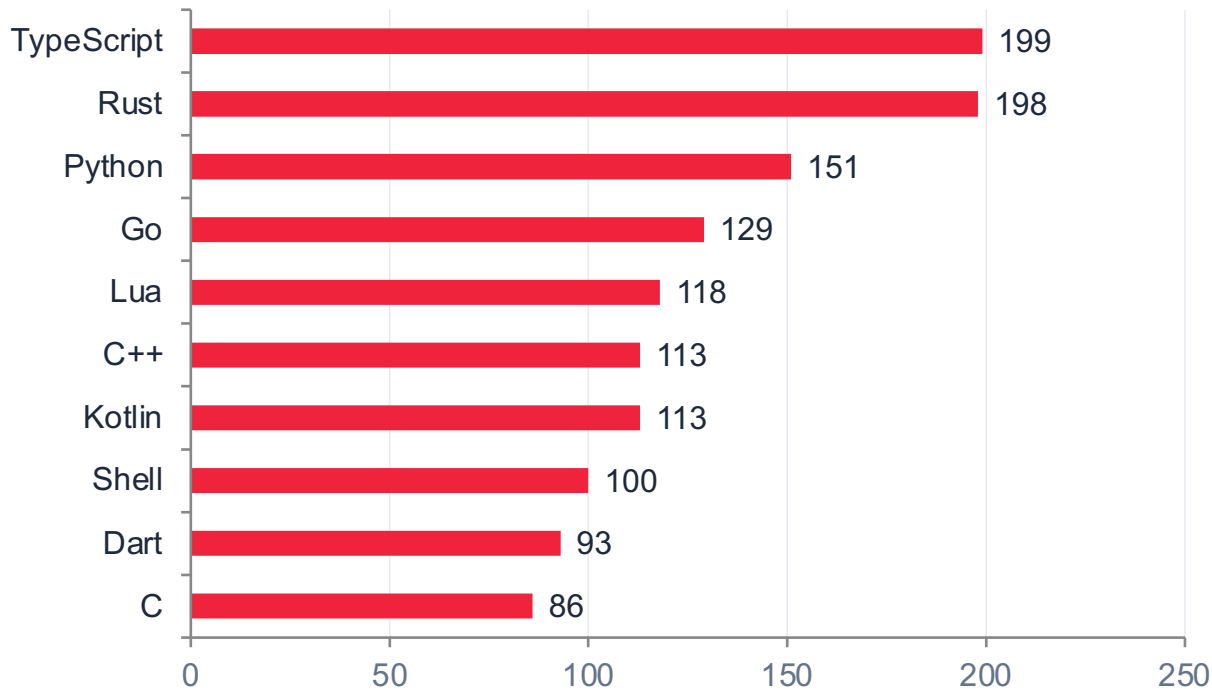
Developers show strong loyalty to their chosen ecosystems.

Adoption Intent is Modest:

- Python: 6%
- TypeScript: 5%
- Java: 4%
- SQL: 4%

All adoption intent sits below 6%, highlighting 'comfort zone effect' where

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198



Why They Lead

TypeScript (199) and Rust (198) score nearly identically as the most loved languages.

Both are prized for:

- Compiler safety guarantees
- Modern developer experience
- Expressive type systems

High satisfaction reduces
turnover and boosts

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Application Domain	Top Language	Share
User Interface	TypeScript	62%
Application Logic & Workflows	Java	55%
APIs & Services	Java	49%
Integrating with APIs	Python	38%
AI / Machine Learning	Python	31%
Data Science & Analytics	Python	28%
System Tools & Components	C++	27%

Domain Supremacy

Language choice is driven by domain, not general preference.

AI & Data:

Python's growth correlates directly with the AI/ML boom.

Frontend UI:

TypeScript (62%) and JavaScript (58%) own the

What This Means for Your Technology Strategy in 2025

- Invest in Python for AI/ML teams

- TypeScript is the future of frontend and full-stack.

- Rust and Go are the growth plays

- PHP and Ruby hiring pools are shrinking

- Developer satisfaction correlates with type safety

Factor modern tooling into your retention strategy to keep top talent.